

# Individual-Assignment-5

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## Set up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

### Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

## Class Code

```

# ---- sites object ----

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# ---- cleaning taxon_list ----

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# ---- cleaning water_quality ----

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames

```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# ---- cleaning aq_ins ----

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)

```

```

summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site (ordered by median salinity, descending)
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
  site_full = fct_rev(site_full))

# ---- salinity plot ----

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent medians
stat_summary(fun = median,
  color = "red",
  size = 1) +

```

```

# relabelling x- and y-axes
labs(x = "Site",
      y = "Median salinity (ppt)",
      title = "Site salinity") +
# different theme
theme_bw()

# ---- copepods plot ----

# creating new object from clean aquatic inverts
copepods_df <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods_df,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Copepods") +
# different theme
theme_bw()

```

# 1. Visualizing differences across sites in major taxon abundance

## a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

## b. Wrangle your data

```
# creating new object from clean aquatic inverts for ostracods
ostracods_df <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows to check output
slice_sample(ostracods_df, n = 10)
```

```
# A tibble: 10 x 24
```

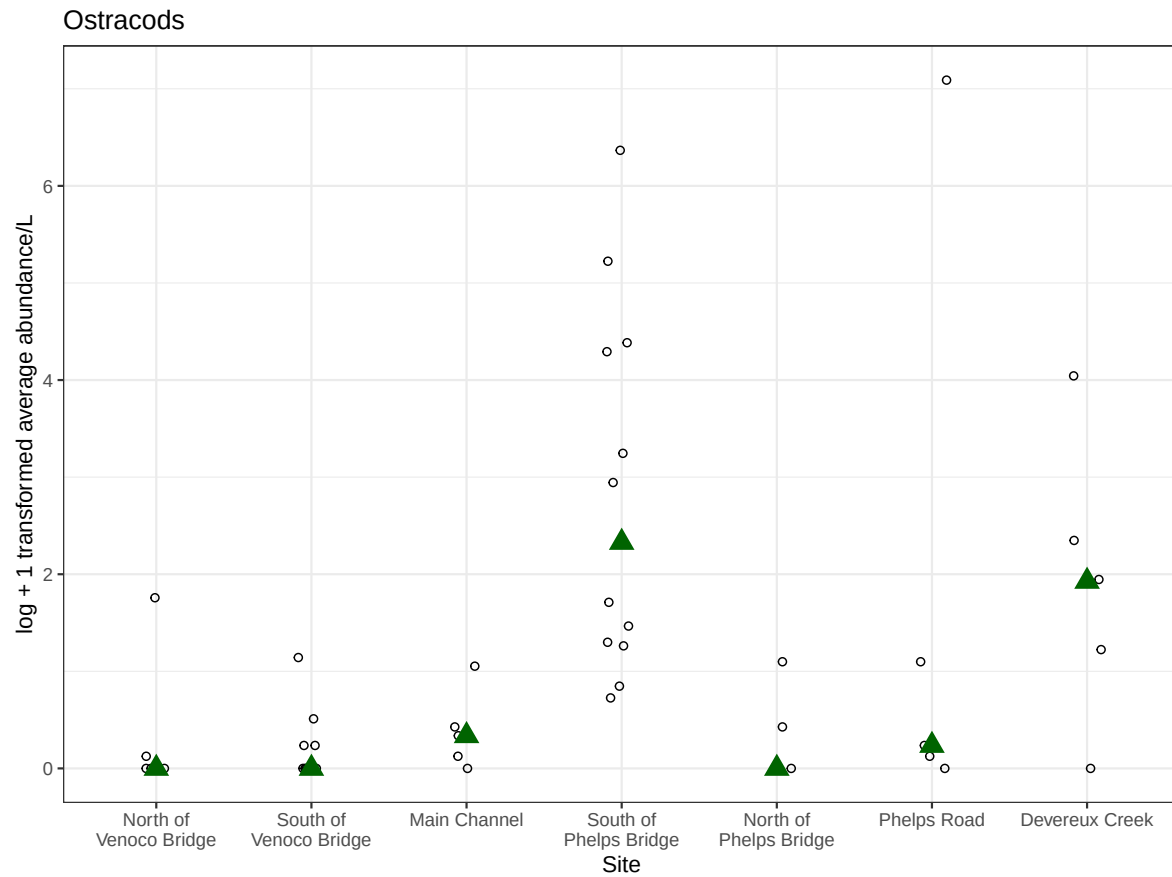
|    | date_site       | date                | site  | taxon | ave_lit | med_ph | med_sal | med_do |
|----|-----------------|---------------------|-------|-------|---------|--------|---------|--------|
|    | <chr>           | <dtm>               | <chr> | <chr> | <dbl>   | <dbl>  | <dbl>   | <dbl>  |
| 1  | 2022-08-12_NVBR | 2022-08-12 00:00:00 | NVBR  | ostr~ | 0       | NA     | NA      | NA     |
| 2  | 2022-01-24_NEC  | 2022-01-24 00:00:00 | NEC   | ostr~ | 0.267   | NA     | NA      | NA     |
| 3  | 2022-02-25_NEC  | 2022-02-25 00:00:00 | NEC   | ostr~ | 0       | 8.75   | 44.1    | 9.87   |
| 4  | 2023-02-28_NPB1 | 2023-02-28 00:00:00 | NPB1  | ostr~ | 0       | NA     | NA      | NA     |
| 5  | 2023-02-21_NEC  | 2023-02-21 00:00:00 | NEC   | ostr~ | 0       | 8.56   | NA      | 8.8    |
| 6  | 2023-08-12_NEC  | 2023-08-12 00:00:00 | NEC   | ostr~ | 0       | NA     | NA      | NA     |
| 7  | 2022-05-13_NVBR | 2022-05-13 00:00:00 | NVBR  | ostr~ | 0       | 9.72   | 74.2    | 11.4   |
| 8  | 2022-02-23_NVBR | 2022-02-23 00:00:00 | NVBR  | ostr~ | 0       | 8.86   | 45.2    | 10.2   |
| 9  | 2022-01-28_NDC  | 2022-01-28 00:00:00 | NDC   | ostr~ | 9.47    | 7.65   | 2.00    | 9.55   |
| 10 | 2022-08-10_NPB  | 2022-08-10 00:00:00 | NPB   | ostr~ | 2.67    | 8.28   | 8.84    | 9.81   |

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

### c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods_df,
                    # x-axis: site
                    # y-axis: log + 1 transformed average abundance/L
                    mapping = aes(x = site_full,
                                   y = log_ave_lit)) +
# first layer: individual average abundance/L measurements
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# second layer: large filled orange triangle for median - distinct color,
# shape, and size from individual points
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "darkgreen",
            shape = 17) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# axis labels and title indicating taxonomic group
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# non-default theme
theme_bw()

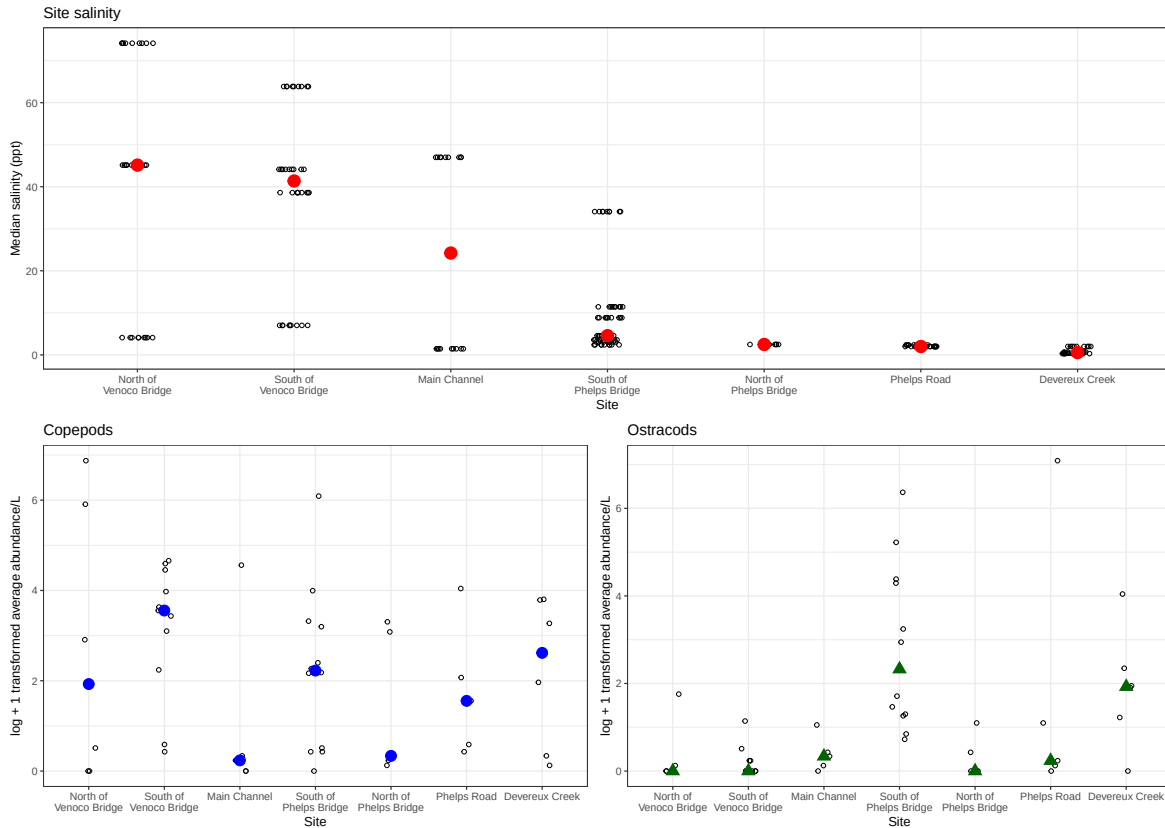
# displaying object
ostracods
```



**d. Create a multipanel figure**

```
# combining salinity, copepods, and ostracods (bottom right) using patchwork
salinity / (copepods | ostracods)
```





### e. Interpret your figure

Copepod median abundance is highest at South of Venoco Bridge with a little over 3 log-units. However the Main Channel and North of Phelps Bridge have medians near 0 log-units. Ostracod median abundance is highest at South of Phelps Bridge with a little over 2 log-units and second highest at Devereux Creek with 2 log-units. However, most other sites showing medians near 0 log-units.

Sites with the highest median salinity such as North and South of Venoco Bridge have ostracod medians near 0, suggesting ostracods are not best suited for high-salinity conditions. The highest ostracod medians occur at South of Phelps Bridge and Devereux Creek, which have lower median salinities (~5 and ~1 ppt), demonstrating a possible preference for water with lower salinity.

## 2. Visualizing total abundance as a function of salinity

### a. Plan your figure

- x-axis: med\_sal
- y-axis: log\_ave\_lit
- color: taxon
- geometry: geom\_point()
- facets: facet\_wrap(~ taxon, scales = "free")

### b. Wrangle your data

```
# creating new object for abundance vs. salinity visualization
sal_abund_df <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon names to title case for readable facet labels
  mutate(taxon = str_to_title(str_replace_all(taxon, "_", " "))) |>
  # reorder taxon factor levels by maximum abundance (descending)
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = "max", .desc = TRUE)) |>
  # keep only columns needed for the plot
  select(taxon, med_sal, log_ave_lit, ave_lit)

# display 10 random rows to check output
slice_sample(sal_abund_df, n = 10)
```

# A tibble: 10 x 4

|    | taxon                       | med_sal | log_ave_lit | ave_lit |
|----|-----------------------------|---------|-------------|---------|
|    | <fct>                       | <dbl>   | <dbl>       | <dbl>   |
| 1  | Nematode                    | NA      | 0           | 0       |
| 2  | Annelida Polychaete         | 1.45    | 0           | 0       |
| 3  | Ostracod                    | NA      | 0.427       | 0.533   |
| 4  | Nematode                    | NA      | 0           | 0       |
| 5  | Nematode                    | 63.8    | 0           | 0       |
| 6  | Annelida Polychaete         | 45.2    | 0           | 0       |
| 7  | Annelida Oligochaete        | 3.08    | 0           | 0       |
| 8  | Copepod                     | NA      | 4.04        | 56      |
| 9  | Hemiptera Corixidae Boatman | 47      | 0.125       | 0.133   |
| 10 | Copepod                     | NA      | 0.588       | 0.8     |

C.